Light

GCSE Computer Science — Character Sets

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Character Sets

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a character set?

- **A.** A collection of binary codes assigned to characters
- **B.** A type of computer font
- **C.** A keyboard layout
- **D.** A text file format

Q2 (1 mark)

What does ASCII stand for?

- **A.** American Standard Code for Information Interchange

- B. Advanced System Code for Internet Integration
- C. Automatic Standard Character Input Interface
- D. American Symbol Code for International Input

Q3 (1 mark)

Standard ASCII uses how many bits to represent each character?

- A. 5 bits
- B. 6 bits
- C. 8 bits
- D. 7 bits

Q4 (1 mark)

Extended ASCII uses how many bits to represent each character?

- A. 6 bits
- B. 7 bits
- C. 16 bits
- D. 8 bits

Q5 (1 mark)

How many different characters can standard 7-bit ASCII represent?

- A. 128
- B. 64
- C. 256
- D. 512

Q6 (1 mark)

How many different characters can 8-bit extended ASCII represent?

- A. 256
- B. 128
- C. 512
- D. 1024

Q7 (1 mark)

Which of the following is NOT included in the standard ASCII character set?

- A. Uppercase letters A-Z
- B. Digits 0-9
- C. Punctuation marks like . and ,
- D. The euro symbol €

Q8 (1 mark)

Why are character sets needed in computing?

- ○ **A.** To allow computers to store and process text as binary data
- ○ **B.** To make text appear in different colours
- ○ **C.** To compress text files
- ○ **D.** To encrypt messages

Q9 (1 mark)

In ASCII, uppercase letters have consecutive codes starting from 65. What type of characters have codes 0-31?

- ○ **A.** Lowercase letters
- ○ **B.** Numbers
- ○ **C.** Punctuation marks
- ○ **D.** Control characters

Q10 (1 mark)

What was the main reason for developing extended ASCII?

- ○ **A.** To make text files smaller
- ○ **B.** To replace standard ASCII completely
- ○ **C.** To speed up text processing

- **D.** To include additional characters like accented letters

Q11 (1 mark)

If a character set uses 6 bits per character, how many different characters can it represent?

- **A.** 32
- **B.** 256
- **C.** 128
- **D.** 64

Q12 (1 mark)

Extended ASCII is designed to be compatible with standard ASCII. What does this mean?

- **A.** They use different codes for the same characters
- **B.** Characters 0-127 have the same codes in both
- **C.** Extended ASCII cannot display standard ASCII text
- **D.** Standard ASCII files are larger than extended ASCII files

Q13 (2 marks)

In ASCII, the uppercase letter 'A' has the code 65.
What is the ASCII code for 'E'?

- **A.** 67

- B. 68
- C. 69
- D. 70

Q14 (2 marks)

In ASCII, 'A' = 65 and 'a' = 97.
What is the difference between uppercase and lowercase letter codes?

- A. 26
- B. 30
- C. 32
- D. 36

Q15 (2 marks)

In ASCII, uppercase 'A' = 65 and lowercase 'a' = 97.
What is the ASCII code for lowercase 'h'?

- A. 72
- B. 102
- C. 104
- D. 136

Q16 (2 marks)

A character set needs to represent exactly 512 different characters.

What is the minimum number of bits required per character?

- ○ **A.** 8 bits
- ○ **B.** 9 bits
- ○ **C.** 10 bits
- ○ **D.** 11 bits

Q17 (2 marks)

The ASCII code for 'W' is 87 in denary.
What is this in binary?

- ○ **A.** 1010101
- ○ **B.** 1010111
- ○ **C.** 1011001
- ○ **D.** 1011011

Q18 (2 marks)

A text file contains 2,000 characters encoded using 7-bit ASCII.
What is the minimum file size in bits?

- ○ **A.** 7,000 bits
- ○ **B.** 14,000 bits
- ○ **C.** 16,000 bits
- ○ **D.** 28,000 bits

Q19 (2 marks)

A document contains 5,000 characters encoded using extended ASCII.
What is the file size in bytes?

- ○ **A.** 4,375 bytes
- ○ **B.** 5,000 bytes
- ○ **C.** 10,000 bytes
- ○ **D.** 40,000 bytes

Q20 (2 marks)

In ASCII, the digit '0' has code 48.
What is the ASCII code for the digit '7'?

- ○ **A.** 7
- ○ **B.** 52
- ○ **C.** 55
- ○ **D.** 57

Q21 (2 marks)

A character set can represent 1,024 different characters.
How many bits does it use per character?

- ○ **A.** 8 bits
- ○ **B.** 9 bits
- ○ **C.** 10 bits

- **D.** 11 bits

Q22 (2 marks)

In ASCII, which sequence of characters has consecutive codes?

- **A.** 'A', 'a', 'B', 'b'
- **B.** '0', '1', '2', '3'
- **C.** 'A', '1', 'B', '2'
- **D.** 'a', 'A', 'b', 'B'

Q23 (2 marks)

Extended ASCII can represent 256 characters.
Why is this still insufficient for global use?

- **A.** 256 characters is too many for computers to process
- **B.** Many languages use thousands of unique characters
- **C.** Extended ASCII uses too much storage space
- **D.** Modern computers cannot read extended ASCII

Q24 (3 marks)

The word 'HIND' is stored using ASCII.
If 'A' = 65, what is the total of all ASCII codes for 'HIND'?

- **A.** 289

- ○ **B.** 291
- ○ **C.** 293
- ○ **D.** 295

Q25 (3 marks)

A new character set needs to include:

- 26 uppercase letters
- 26 lowercase letters
- 10 digits
- 34 special symbols
- 32 control characters

What is the minimum number of bits required per character?

- ○ **A.** 6 bits
- ○ **B.** 7 bits
- ○ **C.** 8 bits
- ○ **D.** 9 bits

Q26 (3 marks)

Document A contains 8,000 characters stored in 7-bit ASCII.
Document B contains 7,000 characters stored in 8-bit extended ASCII.
Which document has the larger file size?

- ○ **A.** Document A is larger
- ○ **B.** Document B is larger
- ○ **C.** They are exactly the same size

- ○ **D.** Cannot be determined without more information

Q27 (3 marks)

A programmer notices that converting uppercase text to lowercase in ASCII can be done by adding 32 to each character code.
If 'CAT' has codes 67, 65, 84, what are the codes for 'cat'?

- ○ **A.** 99, 97, 116
- ○ **B.** 99, 97, 114
- ○ **C.** 97, 99, 116
- ○ **D.** 35, 33, 52

Q28 (3 marks)

A messaging app needs to support a character set with 4,096 different characters.
A message contains 500 characters. What is the file size in bytes?

- ○ **A.** 500 bytes
- ○ **B.** 750 bytes
- ○ **C.** 1,000 bytes
- ○ **D.** 6,000 bytes

Q29 (3 marks)

A company is designing a character set for a specialist application. They need exactly 200 characters.

If they use the minimum number of bits, how much storage space is wasted per character?

- ○ **A.** No space is wasted
- ○ **B.** 56 unused codes per character's worth of bits
- ○ **C.** The unused codes represent about 22% waste
- ○ **D.** 200 bits are wasted in total

Q30 (3 marks)

A database stores customer names using ASCII. The field allows a maximum of 50 characters.
If extended ASCII is used and the field must always use fixed-width storage, what is the storage per name in bytes?

- ○ **A.** 43.75 bytes
- ○ **B.** 50 bytes
- ○ **C.** 100 bytes
- ○ **D.** 400 bytes

Total: **55** marks • Time: **30 min**

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